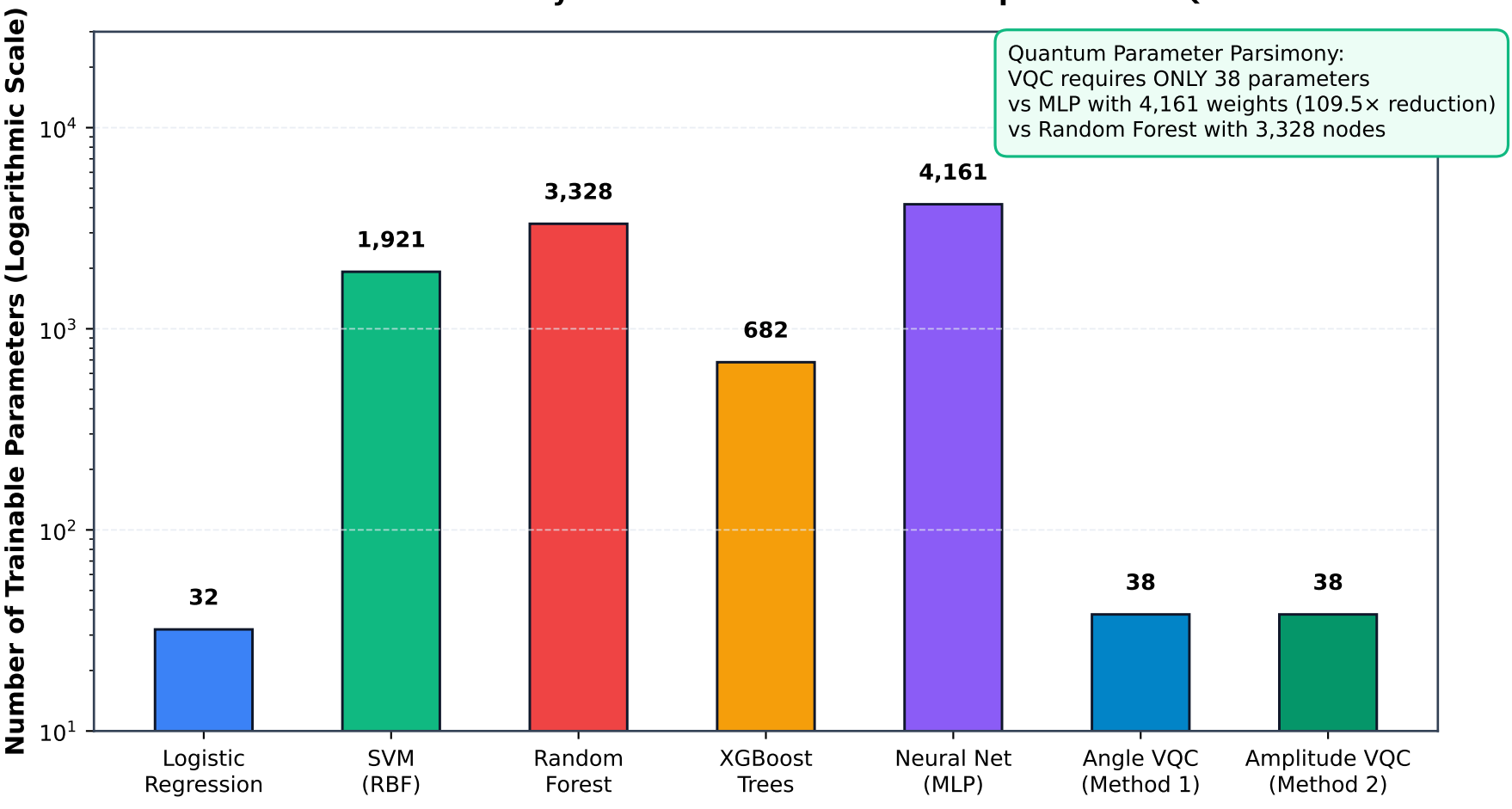
Trainable Parameter Parsimony: Extreme Dimensional Compression in Quantum Circuits
Number of Trainable Parameters (Logarithmic Scale)
$10^4$
$10^3$
$10^2$
$10^1$
Quantum Parameter Parsimony:
VQC requires ONLY 38 parameters
vs MLP with 4,161 weights (109.5× reduction)
vs Random Forest with 3,328 nodes
32
1,921
3,328
682
4,161
38
38
Logistic Regression
SVM (RBF)
Random Forest
XGBoost Trees
Neural Net (MLP)
Angle VQC (Method 1)
Amplitude VQC (Method 2)